

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (EE)

December 2011

EE 203 Electrical Measurement & Measuring Instruments

Date: 08.12.2011, Monday

Time: 10:00 a.m. To 1:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Explain in detail loss of charge method for measurement of high resistance. [05]
- (b) Explain substitution method for measurement of medium resistance. [02]
- Q - 2 (a) Explain in detail Schering bridge [04]
- (b) Explain Maxwell's inductance-capacitance bridge. Also state it's advantages and disadvantages. [06]
- (c) Explain frequency measurement by Wien's bridge. [04]

OR

- Q - 2 (a) Explain Hay's bridge. [04]
- (b) A Maxwell's inductance bridge is shown in Fig.1. Arm ab consists of a coil with inductance L_1 and resistance r_1 in series with a non-inductive resistance R . Arm bc and cd are each a non-inductive resistance of $100\ \Omega$. Arm ad consists of standard variable inductor L_2 of resistance $32.7\ \Omega$. Balance is obtained when $L_2 = 47.8\ \text{mH}$ and $R = 1.36\ \Omega$. Find the resistance and inductance of coil in arm ab. [04]

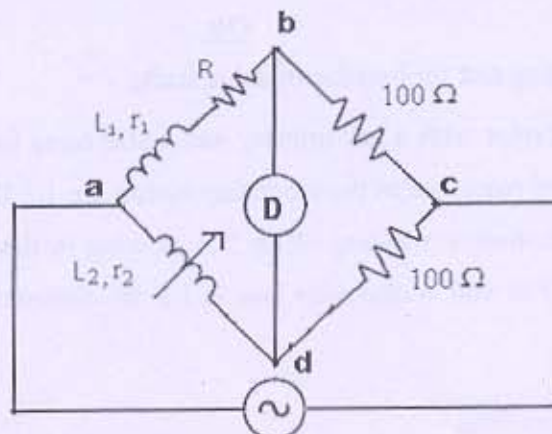


Fig. 1

- (c) Explain with a neat diagram Heaviside mutual inductance bridge. [06]

- Q - 3 (a) Explain briefly different types of A.C. potentiometers. [07]
 (b) Explain different types of static characteristics of measuring instruments [03]
 (c) Explain in detail absolute measurement of current. [04]

OR

- Q - 3 (a) Explain in detail crompton's potentiometer. [07]
 (b) Explain step by step method for determination of B-H curve for a specimen [03]
 (c) Explain in detail absolute measurement of resistance [04]

SECTION - II

- Q - 4 (a) Explain with a neat diagram Nalder-lipman power factor meter. [07]
 Q - 5 (a) Derive general torque equation of a moving iron type instrument. Also state [07]
 advantages and disadvantages of moving iron type instrument.
 (b) Explain briefly operation of electrostatic instruments. Also derive force and torque [07]
 equations of electrostatic instruments.

OR

- Q - 5 (a) Explain with a neat diagram repulsion type Moving Iron instrument [06]
 (b) What are the sources of error in PMMC instrument? [04]
 (c) State and explain advantages and disadvantages of PMMC instrument. [04]
 Q - 6 (a) Draw the phasor diagram of P.T. and derive the equation of transformation ratio and [07]
 phase angle error.
 (b) Explain construction and operation of induction type 1- Φ energy meter. [07]

OR

- Q - 6 (a) Explain Varley loop test for location of cable faults. [07]
 (b) A current transformer with a bar primary has a 300 turns in its secondary winding. [07]
 The resistance and reactance of the secondary circuit are 1.5Ω and 1.0Ω respectively
 including the transformer winding. With 5 A flowing in the secondary winding the
 magnetizing mmf is 100 A and iron loss is 1.2 W. Determine the ratio and phase
 angle errors.
